

PURERUA, New Zealand

WMO: 930230

Lat: 35.125S

Lon: 174.017E

Elev: 285

StdP: 14.55

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.9	44.7	37.0	32.6	46.1	38.9	35.2	47.4	35.7	55.0	30.3	56.8	6.5	200	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.6	75.3	67.4	73.8	66.8	72.5	66.3	70.6	72.6	69.5	71.6	68.4	70.8	8.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.0	111.4	71.5	68.7	106.6	70.8	67.5	102.0	70.0	34.7	72.8	33.7	71.7	32.9	70.9	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	20.9	17.5	40.2	79.3	1.8	2.1	38.9	80.8	37.8	82.0	36.8	83.1	35.5	84.6		
(5)			38.2	72.3	1.6	1.7	37.0	73.6	36.1	74.5	35.2	75.5	34.0	76.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.9	66.0	67.3	65.5	62.4	58.9	55.3	53.8	53.9	55.6	57.3	59.9	63.7	(6)
(7)		DBStd	5.57	2.83	2.67	2.75	3.07	3.57	3.55	3.19	2.88	2.85	2.75	3.06	2.87	(7)
(8)		HDD50	14	0	0	0	0	0	3	6	4	1	0	0	0	(8)
(9)		HDD65	2051	22	7	27	89	192	291	346	343	282	238	154	60	(9)
(10)		CDD50	3642	497	485	482	372	275	162	125	126	169	227	298	424	(10)
(11)		CDD65	202	54	72	44	11	1	0	0	0	0	0	2	18	(11)
(12)		CDH74	116	41	53	12	1	0	0	0	0	0	0	1	8	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.3	7.4	7.5	7.5	7.0	6.9	7.2	7.8	7.6	7.4	7.4	7.3	7.1	(14)
(15)	Precipitation	PrecAvg	57.6	3.7	4.2	4.4	5.2	5.9	6.6	7.0	6.2	4.8	3.9	3.3	4.1	(15)
(16)		PrecMax	73.9	19.3	10.8	13.2	15.0	9.8	13.4	15.4	10.9	12.1	8.3	9.3	9.4	(16)
(17)		PrecMin	43.6	0.4	1.1	0.2	0.2	0.3	2.4	1.3	2.5	1.7	1.6	0.4	0.0	(17)
(18)		PrecStd	10.1	3.9	2.6	3.7	3.4	2.5	2.9	3.4	2.0	2.2	1.7	2.3	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.6	77.9	75.5	72.4	68.6	65.2	63.0	62.9	65.7	68.5	72.4	74.9	(19)
(20)			MCWB	67.3	69.3	66.9	66.5	64.0	62.2	59.3	57.6	58.5	59.4	62.6	64.9	(20)
(21)		2%	DB	75.0	75.5	73.5	70.4	67.2	63.5	61.7	61.4	63.5	65.8	69.4	72.4	(21)
(22)			MCWB	66.8	68.6	66.3	65.3	63.2	60.6	58.0	56.9	57.9	58.8	61.7	64.8	(22)
(23)		5%	DB	73.3	74.1	72.2	69.2	66.1	62.4	60.5	60.3	62.2	64.4	67.6	70.7	(23)
(24)			MCWB	66.9	67.7	65.9	64.6	62.5	59.4	56.8	56.3	57.6	58.5	61.2	64.4	(24)
(25)		10%	DB	71.8	72.7	71.0	68.0	65.0	61.3	59.4	59.3	61.0	63.1	66.1	69.3	(25)
(26)			MCWB	65.9	66.9	65.4	63.6	61.4	58.2	55.8	55.6	56.9	58.0	60.6	63.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.4	72.6	70.9	70.0	66.3	63.7	61.2	60.5	61.2	62.7	66.1	69.4	(27)
(28)			MCDWB	73.2	74.6	72.2	70.7	67.0	64.4	62.2	61.2	62.8	65.0	68.2	71.5	(28)
(29)		2%	WB	69.7	71.1	69.4	67.8	64.9	61.9	59.3	59.1	60.1	61.4	64.6	67.8	(29)
(30)			MCDWB	72.2	73.1	71.2	69.0	66.2	62.8	60.5	60.3	61.7	63.4	66.7	69.6	(30)
(31)		5%	WB	68.5	70.0	68.1	66.3	63.6	60.5	58.0	57.8	59.1	60.5	63.3	66.6	(31)
(32)			MCDWB	71.2	72.2	70.7	68.1	65.4	61.8	59.6	59.4	60.9	62.5	65.6	68.5	(32)
(33)		10%	WB	67.4	68.9	66.8	64.8	62.2	59.0	56.6	56.3	58.1	59.5	62.1	65.5	(33)
(34)			MCDWB	70.4	71.3	69.9	67.0	64.4	60.9	58.7	58.5	60.2	62.0	64.7	67.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.1	10.6	10.9	10.5	10.1	9.7	9.8	10.2	10.7	11.6	11.5	10.7	(35)
(36)			MCDBR	12.7	12.3	11.9	10.8	9.8	8.9	9.6	10.4	11.4	12.7	13.4	12.8	(36)
(37)			MCWBR	7.4	7.1	6.9	6.9	7.0	6.9	7.0	7.2	7.7	8.0	8.2	7.8	(37)
(38)		5% WB	MCDWB	9.4	8.3	8.4	7.4	7.5	6.9	7.2	8.0	8.6	9.5	10.3	8.7	(38)
(39)			MCWBR	6.6	5.8	6.0	6.0	6.2	6.6	6.4	6.8	6.8	7.3	7.6	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.343	0.341	0.329	0.316	0.310	0.304	0.307	0.309	0.332	0.334	0.333	0.341	(40)	
(41)		taud	2.576	2.614	2.637	2.667	2.655	2.654	2.637	2.627	2.533	2.533	2.566	2.587	(41)	
(42)		Ebn at Noon	315	309	301	288	272	265	270	285	295	307	316	317	(42)	
(43)		Edh at Noon	33	31	28	24	22	20	22	25	31	33	34	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2161	1893	1556	1144	823	677	732	998	1353	1741	1993	2063	(44)	
(45)		RadStd	136	129	102	72	45	42	43	56	63	109	136	177	177	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page